

Week 3: Modeling & Sketch of Method

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I. PROJECT TITLE

CVaR-MPPI for Highway Lane-Change Planning.

II. MAIN IDEA

A risk-sensitive lane-change planner that augments Model Predictive Path Integral (MPPI) with a Conditional Value-at-Risk (CVaR) objective. Augmentation of CVaR replaces the risk-neutral expectation over rollouts with a sampled tail-risk backup. The resulting controller suppresses lane-changes that are efficient on average but unsafe under rare traffic outcomes.

III. PROBLEM FORMULATION

Let $x_k \in \mathbb{R}^{n_x}$ be the joint ego-traffic state, $u_k = v_k + \epsilon_k \in \mathbb{R}^{n_u}$ the perturbed ego control with $\epsilon_k \sim \mathcal{N}(0, \Sigma_\epsilon)$, and w_k exogenous traffic noise. The simulated stochastic dynamics $\tilde{x}_{k+1} = \tilde{F}(\tilde{x}_k, u_k, w_k)$ form an unbiased estimate of the nominal $x_{k+1} = F(x_k, u_k)$ [5]. Over a horizon K with state cost q , terminal cost ϕ , and risk cost $L(\tilde{x}) = \sum_{k=0}^{K-1} \ell(\tilde{x}_k)$, we seek

$$\min_v \mathbb{E} \left[\phi(x_K) + \sum_{k=0}^{K-1} \left(q(x_k) + \frac{\lambda}{2} v_k^\top \Sigma_\epsilon^{-1} v_k \right) \right] \quad (1)$$

s.t. $\text{CVaR}_\alpha(L(\tilde{x})) \leq C_u,$

where $\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]$ is the conditional expectation of the worst $1 - \alpha$ tail [2], $\alpha \in (0, 1)$ the confidence level, $\lambda > 0$ the MPPI temperature, and C_u caps the admissible tail risk.

IV. SKETCH OF METHOD

A. Offline Component

The offline stage instantiates the dynamic-programming foundation of CVaR control by Chow et al. [4]. Augmenting the state with a confidence variable $y \in (0, 1]$, the CVaR-MDP value $V^*(x, y) = \min_{\mu \in \Pi_H} \text{CVaR}_y(\sum_t \gamma^t Z_t \mid x_0 = x, \mu)$ admits a γ -contractive, concavity-preserving Bellman recursion

$$\mathbf{T}[V](x, y) = \min_a [c(x, a) + \gamma \max_{\xi \in \mathcal{U}_{\text{CVaR}}(y, P)} \sum_{x'} \xi(x') V(x', y \xi(x')) P(x' \mid x, a)] \quad (2)$$

[4]. We discretize a coarse three-lane MDP over (lane index, gap-to-lead, ego speed) on $\sim 10^3$ cells with 21 log-spaced confidence points and run interpolated value iteration

(Algorithm 1 of [4]) to obtain $V^*(x, y)$, which is loaded as the MPPI terminal cost $\phi(x_K) \leftarrow V^*(\Pi_{x_K}, y)$, where Π_{x_K} maps the continuous terminal state to the coarse MDP grid. This provides a tail-risk-shaped cost-to-go beyond the rollout horizon. Offline complexity is $\mathcal{O}(|\mathcal{X}| |\mathcal{Y}| |\mathcal{A}| C_\zeta n_{\text{iter}})$ where C_ζ is the cost of the inner risk-envelope optimization. This computational cost is paid once before deployment; runtime queries are linear-interpolation lookups in microseconds.

B. Online Component

The online stage executes a Risk-Aware MPPI receding-horizon controller [1], [5]. At each step we sample M control sequences $u_k^m = v_k + \epsilon_k^m$, propagate K -step rollouts under the nominal bicycle to accumulate S_m , and for each m simulate N stochastic traffic futures via $\tilde{F}$ to compute the Monte-Carlo CVaR estimate

$$\widehat{\text{CVaR}}_\beta^m = \frac{1}{N_o} \sum_{n=0}^{N-1} \mathbf{1}_{\{L(\tilde{x}^{m,n}) \geq \widehat{\text{VaR}}_\beta^m\}} L(\tilde{x}^{m,n}), \beta \in (0, 1) \quad (3)$$

empirical tail-average CVaR estimator used in RA-MPPI. [5]. The penalty is folded into a modified rollout cost $\tilde{S}_m = \widehat{\text{ACVaR}}_\beta^m \mathbf{1}_{\{\widehat{\text{CVaR}}_\beta^m > C_u\}} + S_m$, importance weights $\tilde{\omega}_m = \exp(-(\tilde{S}_m - \min_m \tilde{S}_m)/\lambda)$ are formed, and the mean control is updated as $v^+ = \sum_m \tilde{\omega}_m u^m / \sum_m \tilde{\omega}_m$; the first input v_0^+ is applied and $v \leftarrow v^+$ warm-starts the next step. This stage is suitable because GPU-parallel rollouts handle the continuous nonlinear bicycle within the ~ 100 ms control budget, and Monte-Carlo CVaR accommodates arbitrary, non-Gaussian traffic disturbances. It complements the offline stage by injecting fine-grained interaction with current surrounding vehicles that the coarse V^* cannot resolve, while V^* supplies the long-horizon CVaR-Bellman backup that a finite- K rollout would otherwise miss.

V. WORKING EXAMPLE MODELING

A. Modeling & Simulation

The ego vehicle uses a 4-state kinematic bicycle, $\dot{X} = v \cos \psi$, $\dot{Y} = v \sin \psi$, $\dot{\psi} = \frac{v}{L} \tan \delta$, $\dot{v} = a$, with controls $u = (a, \delta)$ and wheelbase L . The environment is a three-lane straight highway populated by 4–10 surrounding vehicles whose longitudinal motion follows a stochastic Intelligent Driver Model with Gaussian acceleration noise $w_k \sim \mathcal{N}(0, \sigma^2)$ and a 2% random lane-change probability per step, providing the rare-but-significant tail events that motivate a CVaR objective. The state cost $q(x)$ combines a smooth collision potential, a lane-tracking term, a desired-speed term, and an effort penalty; the running risk cost

$\ell(\tilde{x}_k)$ is a time-to-collision-shaped potential plus a lateral-encroachment indicator, so high-CVaR rollouts coincide with the near-miss outcomes the planner must avoid.

B. Plan for Implementation

The simulator and controller are implemented in Python with JAX-vmap for vectorised GPU rollouts (target $M = 1024$, $N = 64$, $K = 30$, $\Delta t = 50$ ms); offline value iteration uses NumPy with the linear-interpolation Bellman operator of [4]. Experiments compare standard MPPI against CVaR-MPPI across (i) low/medium/high traffic densities, (ii) hard-brake and cut-in near-miss scenarios, and (iii) a sweep over $\alpha \in [0.7, 0.99]$ and C_u , reporting collision rate, near-miss rate ($\text{TTC} < 1.5$ s), lane-change completion time, and control smoothness to test whether CVaR-MPPI strictly reduces tail-risk events without measurable loss of nominal performance.

REFERENCES

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